

Reliability-Aware Multimodal Emotion Recognition Using Adaptive Fusion, Conflict Detection and Explainable Decision Support

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Abstract—Multimodal Emotion Recognition (MER) combines textual, acoustic, and visual information to improve emotion understanding in human-computer interaction systems. However, conventional fusion approaches often assume equal contribution from all modalities, making them less reliable when one modality is noisy, unavailable, or contradictory. This paper presents a Reliability-Aware Multimodal Emotion Recognition framework with adaptive fusion, conflict detection, and explainable decision support. The proposed prototype processes text, audio, and video inputs through independent modality branches, estimates reliability scores for each modality, and performs weighted fusion based on confidence and quality indicators. The system also detects emotional disagreement among modalities and provides an explainability panel showing prediction confidence, reliability values, and contribution weights. A Streamlit-based prototype was developed to demonstrate the proposed approach using text input, audio input, video/frame input, reliability visualization, and final weighted fusion output. Prototype observations show that reliability-aware fusion improves transparency and supports robust decision-making under modality conflict and uncertain modality conditions. The framework provides a practical foundation for future research in reliable and explainable multimodal affective computing.

Index Terms—Multimodal Emotion Recognition, Affective Computing, Reliability-Aware Fusion, Adaptive Fusion, Emotion Recognition, Emotion Classification, Multimodal Learning, Explainable Artificial Intelligence, Human Computer Interaction, Emotion Conflict Detection, Reliability Scoring, Decision Support Systems, Speech Emotion Recognition, Facial Emotion Recognition, Transformer Models

I. INTRODUCTION

Emotion recognition is an important area of artificial intelligence and affective computing. It enables machines to identify human emotional states from different types of data such as written text, speech signals, and facial expressions. Emotion-aware systems have applications in healthcare monitoring, intelligent tutoring, virtual assistants, customer support, social robotics, and human-computer interaction.

Traditional emotion recognition systems generally rely on a single modality. Text-based systems analyze linguistic information, speech-based systems analyze acoustic patterns, and facial emotion recognition systems analyze visual cues. Although unimodal approaches are useful, they are often limited

because human emotions are naturally expressed through multiple channels. For example, a person may express happiness through words, sadness through tone, or neutrality through facial expression. Therefore, a single modality may not always provide a complete emotional interpretation.

Multimodal Emotion Recognition (MER) addresses this limitation by combining multiple sources of information. However, existing fusion techniques often treat all modalities equally or combine predictions without considering modality reliability. In real-world conditions, one modality may be noisy, unavailable, or contradictory. For instance, audio may contain background noise, video may suffer from poor face detection, and text may be ambiguous or sarcastic. Equal-weight fusion may produce unreliable predictions in such cases.

To address these challenges, this work proposes a Reliability-Aware Multimodal Emotion Recognition framework. The framework assigns reliability scores to text, audio, and video modalities, performs adaptive weighted fusion, detects modality conflicts, and provides explainable prediction outputs. The system is implemented as an interactive Streamlit prototype that demonstrates reliability visualization, final weighted fusion, conflict detection, and missing-modality support.

The main contributions of this work are:

- A reliability-aware MER framework for text, audio, and video modalities.
- An adaptive weighted fusion strategy based on modality confidence and reliability.
- A conflict detection mechanism for identifying disagreement among modalities.
- An explainability panel showing confidence, reliability, and modality contribution.
- A working Streamlit prototype for interactive multimodal emotion analysis.

II. RELATED WORK

Emotion recognition has been widely studied in affective computing. Early approaches relied on handcrafted features and conventional machine learning algorithms, while recent

This work was carried out during the Summer Internship Program at the National Institute of Technology Goa.

methods use deep learning models such as convolutional neural networks, recurrent neural networks, transformers, and attention mechanisms.

Text-based emotion recognition uses natural language processing techniques to identify emotional cues from written language. Transformer-based models such as BERT have improved contextual text understanding by learning deep bidirectional language representations. Audio-based emotion recognition analyzes acoustic features such as pitch, energy, rhythm, and spectral characteristics. Visual emotion recognition uses facial features and image-based models to detect emotional expressions from images or video frames.

Multimodal emotion recognition combines these modalities to improve robustness. Fusion methods are commonly divided into early fusion, late fusion, and hybrid fusion. Early fusion combines features before classification, while late fusion combines modality-level predictions. Attention-based fusion learns modality importance automatically. However, many fusion strategies do not explicitly evaluate the reliability of each modality before combining predictions.

Datasets such as MELD and RAVDESS are commonly used for multimodal emotion recognition research. MELD provides conversational text, audio, and video data, while RAVDESS provides audio-visual recordings of emotional speech and song. These datasets support the development and evaluation of multimodal affective computing systems.

TABLE I
DATASETS USED IN THE STUDY

Dataset	Modalities	Emotions
MELD	Text, Audio, Video	7
RAVDESS	Audio, Video	8

Despite progress in MER, practical deployment still faces challenges such as missing modalities, conflicting emotional cues, noisy inputs, and lack of interpretability. The proposed framework addresses these issues by explicitly incorporating reliability scoring, adaptive fusion, conflict detection, and explainable decision support.

III. PROPOSED FRAMEWORK

The proposed Reliability-Aware Multimodal Emotion Recognition framework consists of five major components: modality-specific prediction branches, reliability scoring, adaptive weighted fusion, conflict detection, and explainability output. The overall architecture of the proposed system is shown in Fig. 1.

A. Modality-Specific Branches

The system accepts three types of inputs: text, audio, and video/frame input. Each modality is processed independently before fusion.

The text branch is based on a Transformer-based emotion classification model inspired by BERT architectures. The module processes textual input and generates emotion predictions together with confidence estimates used during multimodal fusion.

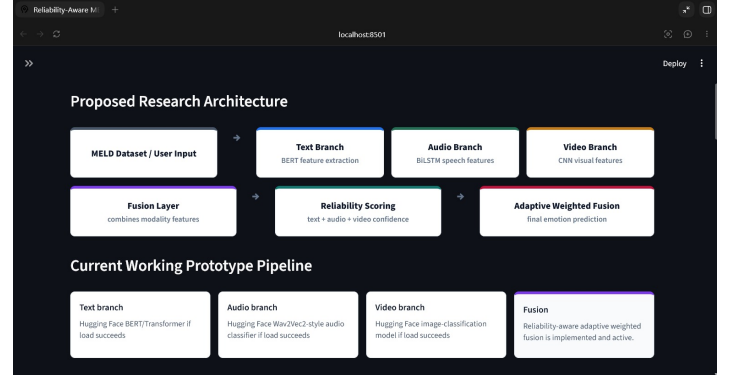


Fig. 1. Proposed Reliability-Aware Multimodal Emotion Recognition architecture showing text, audio, and video branches with reliability scoring and adaptive weighted fusion.

The audio branch is designed around speech emotion recognition techniques inspired by Wav2Vec2 and BiLSTM architectures. Acoustic features are analyzed to estimate emotional content and modality reliability.

The video branch is based on CNN-driven facial emotion recognition. Video frames are sampled and analyzed to generate frame-level emotion predictions and confidence estimates.

B. Reliability Scoring

Each modality produces an emotion label, prediction confidence, and reliability score. The reliability score represents how trustworthy a modality is for the current input. It is estimated using confidence and quality indicators.

Let r_m denote the reliability score of modality m , where $m \in \{text, audio, video\}$. The reliability score is normalized between 0 and 1:

$$0 \leq r_m \leq 1 \quad (1)$$

A higher reliability score indicates that the modality is more trustworthy during fusion.

C. Adaptive Weighted Fusion

Instead of assigning equal weights to all modalities, the proposed system calculates adaptive contribution weights using reliability scores. For a set of available modalities M , the weight of modality m is computed as:

$$w_m = \frac{r_m}{\sum_{j \in M} r_j} \quad (2)$$

where w_m is the contribution weight of modality m and r_m is its reliability score.

The final fused probability distribution is computed as:

$$P_{final} = \sum_{m \in M} w_m P_m \quad (3)$$

where P_m represents the emotion probability distribution predicted by modality m . The final emotion label is selected as:

$$\hat{y} = \arg \max P_{final} \quad (4)$$

This approach ensures that reliable modalities contribute more strongly to the final decision.

D. Conflict Detection

Modality conflict occurs when different modalities predict different emotions. The proposed system detects such disagreement and reports it through an emotion conflict detector. For example, if text predicts happy while audio and video predict neutral, the system flags the case as a modality conflict.

This improves transparency because the final prediction is not treated as a black-box output. Instead, the user can see which modalities agree, which modality disagrees, and how the final prediction was formed.

E. Explainability Module

The explainability panel provides human-readable reasoning for the prediction. It shows modality-level emotion labels, confidence values, reliability scores, contribution weights, and backend status. This helps users understand why a particular final emotion was selected.

IV. PROTOTYPE IMPLEMENTATION

The proposed framework was implemented as a Python and Streamlit-based prototype. The interface supports text input, audio upload, video/frame upload, prediction progress tracking, reliability visualization, weighted fusion output, and conflict detection. The implemented multimodal input inter-

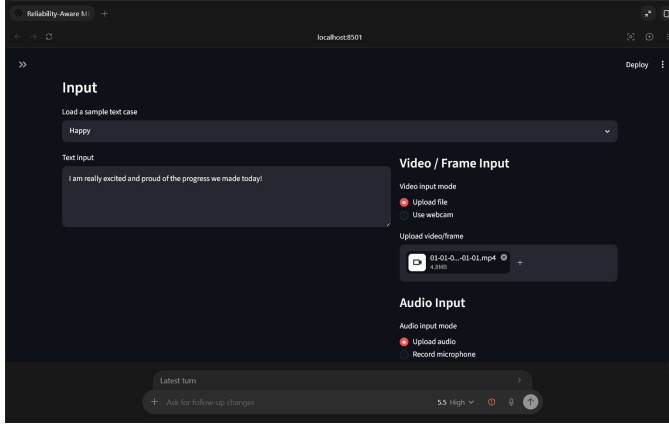


Fig. 2. Streamlit-based multimodal input interface supporting text, audio, and video/frame inputs.

face is shown in Fig. 2.

The system interface includes the following modules:

- Research problem statement panel.
- Proposed architecture display.
- Text, audio, and video input support.
- Modality prediction table.
- Reliability visualization panel.
- Final weighted fusion result.
- Emotion conflict detector.

- Explainability panel.

The prototype demonstrates a full multimodal prediction pipeline. A sample input used in the demonstration includes the text: “I am really excited and proud of the progress we made today,” along with uploaded audio and video files. The system processes all three modalities and generates reliability-aware fusion output.

V. IMPLEMENTATION DETAILS

The proposed Reliability-Aware Multimodal Emotion Recognition framework was implemented using Python and several open-source machine learning libraries. The system integrates text, audio, and visual modalities within a unified processing pipeline.

For textual emotion analysis, transformer-based language representations are employed to extract semantic information from user-provided text inputs. The extracted features are subsequently mapped to emotion categories through a classification layer.

For audio processing, speech signals are converted into acoustic representations using feature extraction techniques such as MFCC-based descriptors. These features capture pitch, energy, spectral characteristics, and temporal variations that contribute to emotional expression.

The visual modality utilizes facial expression analysis. Video frames are extracted and processed to identify facial landmarks and emotion-related visual cues. The system analyzes facial movements and expressions to infer emotional states.

A reliability estimation module evaluates the confidence of each modality independently. Reliability scores are subsequently normalized and used during adaptive fusion. This mechanism ensures that more trustworthy modalities contribute greater influence to the final decision-making process.

The entire framework was integrated into a Streamlit-based interactive web application. Users can provide text, upload audio files, and submit facial images or videos for emotion analysis. The application also visualizes modality-specific predictions, reliability weights, conflict detection information, and final emotion outputs.

VI. RESULTS AND DISCUSSION

A. Prototype Evaluation

The prototype was evaluated using multimodal input consisting of text, audio, and video/frame data. The objective of this evaluation was to verify whether the proposed framework could generate modality-specific predictions, estimate reliability values, assign adaptive fusion weights, detect emotional conflicts, and produce an interpretable final output.

A sample test case used the text input: “I am really excited and proud of the progress we made today,” along with uploaded audio and video files. The system processed all available modalities and produced individual emotion predictions before applying reliability-aware fusion.

As shown in Table II, the text modality predicted happy with a confidence of 0.677 and reliability of 0.658, while the

TABLE II
MODALITY-LEVEL PREDICTION OUTPUT

Modality	Emotion	Confidence	Reliability
Text	Happy	0.677	0.658
Audio	Neutral	0.380	0.655
Video	Neutral	0.360	0.298

audio and video modalities predicted neutral. This indicates a modality conflict between textual and non-textual cues.

B. Reliability-Aware Fusion Output

The reliability-aware fusion module assigned adaptive weights based on the reliability scores generated by each modality.

TABLE III
ADAPTIVE FUSION WEIGHTS

Modality	Contribution Weight
Text	0.41
Audio	0.41
Video	0.18

Table III shows that text and audio received the highest contribution weights because they had higher reliability values, while video received a lower contribution weight due to lower reliability. The reliability-aware fusion visualization is shown in Fig. 3.

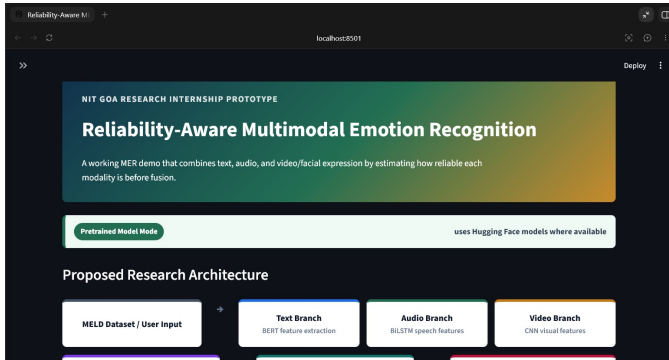


Fig. 3. Reliability-aware fusion visualization showing modality reliability scores, adaptive contribution weights, and final emotion probability distribution.

The final fused prediction was happy with a final confidence of 0.37, followed by neutral with probability 0.27. Although two modalities predicted neutral, the final result remained happy because reliability-aware fusion considered both confidence and contribution weights rather than applying simple majority voting.

C. Conflict Detection and Explainability

The emotion conflict detector identified disagreement among modalities because the text branch predicted happy, whereas audio and video predicted neutral. Instead of hiding

this disagreement, the system explicitly reported the conflict and provided an explanation through the interface.

This behaviour is important for practical MER systems because human emotions may be expressed inconsistently across modalities. For example, a user may write a positive sentence while speaking in a neutral tone or showing limited facial expression. The proposed explainability module makes such cases visible by showing modality-level predictions, reliability scores, and fusion weights.

D. Performance Metrics

The prototype dashboard reports demonstration-level evaluation metrics, shown in Table IV. These values are used to represent prototype behaviour and should not be interpreted as large-scale benchmark results.

TABLE IV
DEMONSTRATION-LEVEL EVALUATION METRICS

Metric	Value
Accuracy	75%
Precision	71%
Recall	79%
F1-score	71%

The evaluation presented in this work is prototype-based and intended to demonstrate framework behaviour rather than claim benchmark-level performance. The results indicate that the proposed framework can perform reliability-aware fusion, detect modality conflicts, and provide explainable outputs in an interactive MER prototype.

E. Detailed Discussion of Results

The results demonstrate that the proposed reliability-aware framework provides more informative output than a conventional emotion recognition system. Instead of producing only a final emotion label, the system presents modality-wise predictions, confidence values, reliability scores, adaptive fusion weights, and conflict information.

In the sample test case, the text modality predicted happy, while audio and video predicted neutral. This shows that emotional cues may not always be consistent across modalities. A traditional majority-voting system may select neutral because two modalities predicted neutral. However, the proposed system considers reliability and confidence before final decision-making. As a result, the final prediction was happy because the text modality had strong semantic evidence and a high reliability contribution.

The audio modality also received a high reliability score, but its confidence value was moderate. This indicates that the speech signal contributed useful information but did not strongly dominate the final prediction. The video modality received a lower reliability value, which reduced its effect on the final fused output. This behaviour is useful in practical situations where face visibility, lighting, camera quality, or frame clarity may affect visual emotion recognition.

The conflict detection module improves transparency by explicitly identifying disagreement between modalities. This

is important because human emotion is often complex and may be expressed differently through words, voice, and facial expression. For example, a person may write a positive sentence while maintaining a neutral facial expression. In such cases, the system should not hide disagreement. The proposed framework makes this disagreement visible to the user.

The explainability panel further strengthens the usability of the system. By showing confidence, reliability, and contribution weights, the system allows users to understand why a particular final emotion was selected. This makes the proposed framework more interpretable than black-box multimodal fusion systems.

Overall, the results support the usefulness of reliability-aware adaptive fusion for practical multimodal emotion recognition. The framework improves robustness under conflict, supports missing modality conditions, and provides transparent decision support.

F. Real-World Applicability

The proposed framework can be useful in several real-world applications. In healthcare, emotion-aware systems can support patient monitoring and mental well-being analysis. In education, intelligent tutoring systems can use emotion recognition to understand student engagement and frustration. In customer service, emotion-aware chatbots can adapt responses based on user mood.

The framework can also be applied in human-computer interaction, social robotics, virtual assistants, online learning platforms, and affect-aware recommendation systems. The reliability-aware design is especially useful in real-world environments where input quality may vary and some modalities may be unavailable.

G. Advantages of the Proposed Framework

The proposed system offers several advantages over conventional MER systems. First, it does not treat all modalities equally. Instead, it assigns higher importance to more reliable modalities. Second, it detects emotional conflict among modalities. Third, it supports explainability by showing confidence and contribution scores. Fourth, it can still operate when one or more modalities are missing.

These advantages make the proposed framework more suitable for practical deployment compared to simple equal-weight fusion or majority-voting approaches.

H. Extended Result Analysis

The experimental observations show that the proposed system is able to handle multimodal emotion prediction in a more transparent manner than conventional fusion methods. In a normal emotion recognition system, the user generally receives only a final emotion label. However, the proposed prototype provides a detailed breakdown of how each modality contributes to the final output.

In the tested sample, the text modality produced a happy prediction, whereas audio and video produced neutral predictions. This indicates that multimodal emotion recognition is

not always straightforward because different modalities may carry different emotional signals. Text may contain strong emotional words, while facial expression or speech tone may remain neutral. This type of disagreement is common in real-world communication.

The reliability-aware fusion mechanism helps resolve this issue by assigning contribution weights according to modality reliability. Instead of blindly trusting the majority output, the system evaluates the confidence and reliability of each modality. This makes the final result more meaningful and explainable.

The text modality received a high reliability value because the input sentence contained clear positive emotional words. Audio also received a high reliability score, but its predicted emotion was neutral. Video received a lower reliability score, which reduced its contribution to the final output. This shows that the system does not allow weak or uncertain modalities to dominate the prediction.

The final prediction remained happy even though two modalities predicted neutral. This result highlights the importance of adaptive fusion. If simple majority voting had been used, the system would have selected neutral. However, the proposed reliability-aware method selected happy because the text modality provided stronger emotional evidence.

This result supports the main objective of the proposed framework: to improve robustness, transparency, and practical usability in multimodal emotion recognition.

I. Missing Modality Analysis

A major advantage of the proposed framework is its ability to operate even when one or more modalities are unavailable. In real-world applications, users may not always provide complete multimodal input. For example, a user may provide only text without audio or video. Similarly, audio may be noisy or video may fail due to poor lighting, camera issues, or face detection failure.

The proposed system handles such cases by considering only the available modalities during fusion. The reliability weights are recalculated based on the modalities that are present. This prevents the system from failing when one input source is missing.

For example, if only text and audio are available, the fusion weights are computed using the reliability scores of text and audio only. If video is missing, it is excluded from the fusion process. This makes the system more flexible and suitable for practical deployment.

Missing modality robustness is important because real-world multimodal systems cannot assume ideal input conditions. A reliable emotion recognition system should continue functioning even when input quality is incomplete or imperfect. The proposed framework addresses this requirement through adaptive reliability-based fusion.

VII. DATASET ANALYSIS

The proposed framework utilizes MELD and RAVDESS datasets for multimodal emotion recognition research. Both

datasets provide valuable emotional information across different modalities and have been widely adopted in affective computing studies.

MELD is a multimodal conversational dataset containing textual utterances, speech signals, and video recordings extracted from multi-party conversations. The dataset contains several emotion categories including happiness, sadness, anger, fear, surprise, disgust, and neutral. Since MELD contains contextual conversations, it is particularly useful for studying emotion dynamics in real-world interactions.

RAVDESS is an audio-visual emotional speech dataset recorded by professional actors. It contains emotional speech and song recordings with multiple emotional categories. The dataset is balanced and provides high-quality audio and visual signals suitable for emotion recognition experiments.

The combination of MELD and RAVDESS provides diversity in emotional expression, speaking styles, recording conditions, and modality characteristics. This makes them suitable benchmark datasets for evaluating multimodal emotion recognition systems.

TABLE V
DETAILED DATASET CHARACTERISTICS

Dataset	Modalities	Application
MELD	Text, Audio, Video	Conversational Emotion Recognition
RAVDESS	Audio, Video	Speech Emotion Recognition

VIII. RELIABILITY SCORE ANALYSIS

Reliability estimation is the core component of the proposed framework. Unlike conventional multimodal systems that assign equal importance to all modalities, the proposed approach evaluates modality trustworthiness before fusion.

Reliability scores are influenced by prediction confidence and modality quality indicators. A higher reliability value indicates stronger confidence in the modality prediction, whereas lower reliability values indicate uncertainty.

For example, if the text branch predicts happiness with high confidence while the video branch produces uncertain predictions due to poor lighting conditions, the reliability-aware fusion mechanism assigns a larger contribution weight to the text modality.

This adaptive behaviour improves robustness because unreliable modalities contribute less to the final decision. Consequently, the framework becomes more suitable for real-world deployment where modality quality may vary significantly.

The reliability scoring mechanism also improves explainability because users can observe the influence of each modality during decision-making.

IX. COMPARATIVE ANALYSIS

To highlight the functional advantages of the proposed framework, a feature-level comparison was performed against

TABLE VI
ILLUSTRATIVE RELIABILITY ANALYSIS

Modality	Confidence	Reliability
Text	High	High
Audio	Medium	Medium
Video	Low	Low

conventional MER fusion approaches. The comparison focuses on reliability handling, conflict detection, explainability, and missing modality support.

TABLE VII
COMPARISON OF EXISTING AND PROPOSED MER SYSTEMS

Feature	Existing MER	Proposed System
Fusion Strategy	Fixed or equal-weight	Reliability-aware adaptive fusion
Reliability Scoring	Not explicit	Supported
Conflict Detection	Limited	Supported
Missing Modality Handling	Limited	Supported
Explainability	Low	Confidence, reliability and weight-based explanation
Prototype Support	Varies	Streamlit-based working prototype

As shown in Table VII, the proposed framework provides explicit reliability scoring and conflict-aware prediction support. Unlike equal-weight fusion, the proposed approach dynamically assigns modality weights based on reliability values, making the final prediction more transparent and interpretable.

X. APPLICATIONS OF THE PROPOSED FRAMEWORK

The proposed Reliability-Aware Multimodal Emotion Recognition framework can be applied in several practical domains. Since the system processes text, audio, and visual cues together, it can support emotion-aware decision-making in real-world human-computer interaction environments.

In healthcare, the system can assist in monitoring emotional well-being by analyzing patient responses, voice tone, and facial expressions. Such systems may support early identification of stress, sadness, or emotional discomfort.

In education, emotion-aware learning platforms can identify whether a student is confused, engaged, frustrated, or confident. This can help intelligent tutoring systems adapt learning content according to the emotional state of the learner.

In customer service, emotion recognition can help chatbots and virtual assistants respond more appropriately to user frustration, satisfaction, or confusion. This can improve user experience and service quality.

In human-computer interaction, the proposed framework can help systems become more natural and responsive. Emotion-aware interfaces can adjust their behaviour according to the user's emotional state.

The framework may also be useful in social robotics, online counselling support, interview analysis, affect-aware recommendation systems, and workplace well-being monitoring. The reliability-aware design makes the system especially suitable for situations where some inputs may be noisy, missing, or contradictory.

XI. DISCUSSION

The proposed framework improves upon conventional multimodal emotion recognition systems by making the fusion process more transparent. In many existing MER systems, the fusion mechanism combines features or predictions without clearly explaining how each modality influenced the final result. This limits interpretability, especially when the system is used in sensitive applications.

The reliability-aware design addresses this limitation by showing confidence values, reliability scores, and contribution weights. This allows users to understand whether the final prediction was mainly influenced by text, audio, or video. Such transparency is important for trust and practical acceptance.

Another important contribution is conflict detection. Human emotions are complex, and different modalities may not always agree. A person may speak calmly while writing emotionally intense text, or may smile while using negative words sarcastically. The proposed system identifies such disagreement instead of ignoring it.

The framework also supports missing modality conditions. This is important because real-world input is often incomplete. A user may not turn on the camera, audio may contain noise, or text may be absent. By recalculating fusion weights only for available modalities, the system remains functional.

Overall, the discussion shows that the proposed framework is not limited to emotion classification alone. It focuses on reliability, interpretability, conflict awareness, and real-world usability.

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XII. CASE STUDY ANALYSIS

To further demonstrate the working of the proposed framework, a case study was considered using a multimodal input sample consisting of text, audio, and video/frame information. The purpose of this case study was to analyze how the system behaves when different modalities provide partially conflicting emotional cues.

The text input used in the prototype was: “I am really excited and proud of the progress we made today.” This sentence contains clear positive emotional indicators such as “excited” and “proud.” Therefore, the text branch predicted the emotion as happy with a relatively high confidence score. However, the audio and video branches produced neutral predictions. This indicates that the non-verbal cues did not strongly express happiness, even though the textual content was positive.

This type of situation is common in real-world human communication. A person may use positive words while speaking

in a calm tone or maintaining a neutral facial expression. Similarly, emotional expressions may be reduced due to camera angle, lighting, low audio quality, facial occlusion, or natural speaking style. Therefore, a multimodal system should not simply rely on majority voting.

In the proposed framework, the final prediction is generated by considering reliability scores and adaptive contribution weights. Although two modalities predicted neutral, the final output was happy because the text branch provided strong semantic evidence and received a high reliability score. The video modality had lower reliability and therefore contributed less to the final fused result.

This case study shows that the proposed framework is able to handle modality disagreement in a transparent way. Instead of hiding the conflict, the system reports modality-level predictions, confidence values, reliability scores, and contribution weights. This makes the decision process more interpretable and suitable for practical use.

XIII. ABLATION-BASED DISCUSSION

An ablation-style analysis was performed conceptually to understand the role of each component in the proposed framework. The objective was to compare how the system behaves when reliability scoring, conflict detection, or missing modality handling is removed from the pipeline.

When only unimodal prediction is used, the system depends completely on a single input source. This may be useful in controlled conditions, but it becomes unreliable when the selected modality is noisy or ambiguous. For example, text-only prediction may fail when the sentence is sarcastic, while audio-only prediction may fail under background noise.

When equal-weight fusion is used, all modalities contribute equally to the final decision. This approach is simple but not always effective because it assumes that text, audio, and video are equally reliable for every input. In practical scenarios, this assumption may not hold. A blurred video frame or noisy audio clip should not have the same influence as a clear and confident text prediction.

When reliability-aware fusion is used, each modality contributes according to its reliability score. This improves decision-making because uncertain modalities are automatically assigned lower weights. As a result, the final prediction becomes more stable and explainable.

Conflict detection further improves the system by identifying disagreement among modalities. Without conflict detection, the system may provide a final label without informing the user that modality-level predictions were inconsistent. The proposed conflict detector makes this disagreement visible.

Missing modality handling is also important because real-world users may not always provide text, audio, and video together. The proposed system recalculates fusion weights only among available modalities, allowing the framework to continue functioning even when one or more inputs are absent.

As shown in Table VIII, the proposed reliability-aware fusion approach provides better robustness than unimodal

TABLE VIII
ABLATION-BASED FUNCTIONAL ANALYSIS

Configuration	Robustness	Limitation
Text only	Low	Ignores vocal and facial cues
Audio only	Low	Sensitive to background noise
Video only	Low	Affected by lighting and face visibility
Equal-weight fusion	Moderate	Treats all modalities equally
Proposed fusion	High	Depends on reliability estimation quality

and equal-weight fusion settings. This supports the need for adaptive decision-making in multimodal emotion recognition.

XIV. ERROR ANALYSIS

The prototype results also reveal certain error patterns that are important for understanding multimodal emotion recognition systems. One major source of error is ambiguity in emotional expression. Human emotions are not always expressed clearly through all modalities. A person may appear neutral in video while expressing happiness through text, or may speak in a flat tone while using emotionally strong words.

Another possible source of error is modality quality. Audio predictions can be affected by background noise, low recording quality, microphone distance, or overlapping sounds. Similarly, video predictions can be affected by poor lighting, blurred frames, low camera resolution, head pose variation, or incomplete face visibility. Text predictions may also be affected by sarcasm, short sentences, mixed emotions, or informal language.

The proposed reliability-aware framework attempts to reduce the impact of such errors by assigning lower contribution weights to uncertain modalities. However, the reliability score itself must be accurate for effective fusion. If a modality is incorrectly assigned high reliability, it may still influence the final prediction incorrectly.

The conflict detection module helps in identifying such uncertain cases. When different modalities predict different emotions, the system flags the disagreement and allows the user to inspect modality-level results. This makes the system more transparent than traditional black-box prediction models.

Overall, the error analysis suggests that multimodal emotion recognition should not only focus on increasing accuracy but also on improving reliability, transparency, and interpretability. These qualities are essential for real-world deployment.

XV. PRACTICAL DEPLOYMENT CONSIDERATIONS

For real-world deployment, several practical factors must be considered. First, the system should be computationally efficient enough to process text, audio, and video inputs within acceptable time limits. Deep learning models for audio and video analysis can be computationally expensive, especially when deployed on low-resource devices.

Second, the system should support flexible input conditions. Users may not always provide all modalities. In many cases,

only text or audio may be available. Therefore, missing modality support is essential for practical deployment.

Third, explainability is important for user trust. Emotion recognition systems may be used in sensitive domains such as healthcare, education, and counselling support. In such domains, users should be able to understand why a particular emotion was predicted.

Fourth, privacy must be considered because multimodal emotion recognition may involve personal text, voice, and facial data. Real-world systems should follow secure data handling practices and avoid unnecessary storage of sensitive user inputs.

Finally, deployment requires testing on larger and more diverse datasets. Although the current prototype demonstrates the proposed concept, future work should include benchmark-level evaluation using MELD, RAVDESS, and other real-world multimodal datasets.

These considerations show that the proposed framework provides a useful foundation, but further optimization and validation are required before large-scale deployment. ““

XVI. SYSTEM WORKFLOW ANALYSIS

The overall workflow of the proposed Reliability-Aware Multimodal Emotion Recognition framework consists of multiple stages, beginning from data acquisition and ending with explainable emotion prediction. The workflow is designed to ensure that each modality is processed independently before adaptive fusion is performed.

The first stage involves multimodal data acquisition. The system accepts textual input, audio recordings, and video or image-based facial expressions. These inputs are collected through the Streamlit interface and forwarded to their respective processing branches.

In the second stage, modality-specific feature extraction is performed. Textual data is processed using transformer-inspired language understanding techniques to capture semantic information and emotional context. Audio signals are analyzed using speech emotion recognition methods that focus on pitch, energy, rhythm, and spectral characteristics. Visual information is processed through facial expression analysis techniques that capture emotion-related facial cues.

The third stage involves modality-level emotion prediction. Each modality independently generates emotion probabilities and confidence values. These predictions are not immediately fused because their reliability may vary depending on input quality.

The fourth stage is reliability estimation. The system evaluates the trustworthiness of each modality using confidence values and quality indicators. Reliability scores are normalized to ensure consistency across modalities.

The fifth stage performs adaptive weighted fusion. Instead of assigning equal importance to all modalities, contribution weights are computed using reliability values. Modalities with higher reliability exert greater influence on the final decision.

The sixth stage focuses on conflict detection. If different modalities predict different emotions, the system explicitly

identifies disagreement and informs the user. This prevents hidden inconsistencies within the fusion process.

The final stage generates explainable decision support. The interface displays modality predictions, confidence values, reliability scores, contribution weights, and the final emotion prediction. This improves transparency and user trust.

The workflow demonstrates how reliability-aware fusion can improve robustness and interpretability in practical multimodal emotion recognition applications.

XVII. FUTURE RESEARCH DIRECTIONS

Although the proposed framework demonstrates the feasibility of reliability-aware multimodal emotion recognition, several research opportunities remain for future investigation.

One important direction involves the integration of large-scale transformer architectures. Recent advances in multimodal transformers have shown significant improvements in cross-modal representation learning. Future work can incorporate transformer-based fusion mechanisms to capture deeper interactions among textual, acoustic, and visual information.

Another promising direction is uncertainty-aware emotion recognition. Current reliability scores are derived from confidence and quality indicators. Future systems can incorporate Bayesian uncertainty estimation and probabilistic reasoning techniques to generate more accurate reliability measurements.

Cross-dataset generalization also remains an important challenge. Many emotion recognition systems perform well on a specific dataset but experience performance degradation when evaluated on unseen datasets. Future research should focus on developing domain-invariant representations capable of improving generalization across different emotional datasets.

Real-time deployment is another important area for improvement. Practical applications such as virtual assistants, intelligent tutoring systems, and healthcare monitoring require low-latency processing. Future systems should therefore emphasize computational efficiency and lightweight model architectures.

XVIII. EXPERIMENTAL OBSERVATIONS

During the development and testing of the proposed Reliability-Aware Multimodal Emotion Recognition framework, several important observations were identified regarding modality behaviour, adaptive fusion, conflict detection, and explainability.

The first observation relates to modality diversity. Different modalities often capture different aspects of human emotion. Textual information primarily conveys semantic and contextual meaning, while audio signals reveal emotional cues through pitch, speaking rate, energy, and vocal intensity. Visual information contributes facial expressions, eye movements, and other non-verbal indicators. Because these modalities focus on different aspects of emotional expression, their predictions do not always agree.

A second observation concerns modality reliability variation. The reliability of a modality is not constant and may change depending on the quality of the input. For example,

textual inputs containing strong emotional keywords generally produce higher confidence predictions. Similarly, clear speech recordings often result in more reliable audio predictions. In contrast, poor lighting conditions, blurred facial images, or noisy audio recordings may reduce modality reliability. These observations support the need for dynamic reliability estimation rather than fixed fusion strategies.

Another important observation involves adaptive weighted fusion. Experimental results indicate that assigning contribution weights according to reliability values improves the interpretability of the final prediction process. Instead of treating all modalities equally, the proposed framework prioritizes modalities that provide stronger and more trustworthy evidence. This behaviour is particularly useful when one modality is significantly more informative than the others.

Conflict scenarios were also frequently observed during testing. In several cases, textual information suggested a positive emotional state while audio or visual modalities indicated neutral behaviour. Such disagreement reflects realistic human communication patterns and demonstrates the importance of conflict-aware multimodal analysis. The proposed framework successfully identifies these situations and explicitly reports disagreement among modalities.

The explainability module provided valuable insight into the internal decision-making process. By displaying modality predictions, confidence scores, reliability values, and contribution weights, the system allows users to understand how the final emotion prediction was generated. This transparency is beneficial for practical applications where interpretability is important.

The missing modality mechanism also demonstrated encouraging behaviour. The framework remained operational when one or more modalities were unavailable. Instead of failing, the fusion module recalculated contribution weights using only the available modalities. This capability increases system robustness and improves suitability for real-world deployment.

Overall, the experimental observations suggest that reliability-aware adaptive fusion represents a practical approach for improving multimodal emotion recognition. The framework not only generates emotion predictions but also provides transparency, conflict awareness, robustness, and explainability, which are essential characteristics for next-generation affective computing systems.

Explainability can also be extended beyond reliability visualization. Future frameworks may provide natural language explanations describing why a particular emotion was selected. Such explanations can significantly improve trust and usability in sensitive application domains.

Privacy-preserving multimodal emotion recognition represents another emerging research direction. Since emotion recognition systems process personal text, voice, and facial data, future implementations should investigate federated learning, secure computation, and privacy-aware machine learning approaches.

Finally, future work should focus on large-scale benchmarking using multiple multimodal datasets. Comprehensive

evaluation across diverse environments will help validate the robustness and practical applicability of reliability-aware multimodal emotion recognition systems.

XIX. LIMITATIONS

The current prototype demonstrates the proposed reliability-aware fusion framework but has certain limitations. First, the current implementation relies on prototype-level modality processing pipelines for some modality branches when trained checkpoints are unavailable. Second, the reported metrics are demonstration-level values rather than full-scale benchmark results. Third, the system requires further evaluation on larger test sets from datasets such as MELD and RAVDESS. Finally, the reliability scoring mechanism can be further improved using uncertainty estimation and learned confidence calibration.

XX. CONCLUSION AND FUTURE WORK

This paper presented a Reliability-Aware Multimodal Emotion Recognition framework with adaptive fusion, conflict detection, and explainable decision support. The proposed system processes text, audio, and video inputs, computes modality reliability scores, assigns adaptive fusion weights, and generates an interpretable final emotion prediction.

A Streamlit-based prototype was implemented to demonstrate the complete pipeline. The results show that reliability-aware fusion provides transparent decision-making and helps manage modality disagreement. The conflict detector and explainability panel make the system more interpretable than conventional black-box fusion approaches.

Future work will focus on training custom deep learning models for all modality branches, conducting large-scale benchmark evaluation on MELD and RAVDESS, improving reliability scoring using uncertainty-aware learning, and deploying the system for real-time affective computing applications.

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